ODSC WEST 2026 | HANDS-ON TUTORIAL

A ML Attribution Framework for Measuring Knowledge Graph Quality in GraphRAG

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In this 45-minute session you will score a knowledge graph against a golden question set, turn every query into structural defect signals, and use LightGBM and SHAP to decide which data-quality fix recovers the most answers. Everything runs in one notebook. You can follow along live or watch on screen; there is nothing to configure during the session if you complete the steps below beforehand.

What you need

Item	Details
Laptop	Any operating system, with a current version of Chrome, Firefox, Edge or Safari.
Google account	Free. Needed for Google Colab, the recommended way to run the notebook.
Paid platforms	None. No API keys, cloud credits or GPU are required.
Knowledge	Comfortable reading Python and pandas. Basic idea of a classifier and a metric such as AUC. No prior experience with knowledge graphs, NetworkX or SHAP is needed.

Before the conference: pick one option

Conference Wi-Fi can be unreliable. Please do this at home or in your hotel, and run the notebook once from top to bottom. A full run takes about three minutes.

Option A: Google Colab (recommended)

1. Sign in to your Google account and open the notebook in Colab:

   https://colab.research.google.com/github/Lokeshs-git/kg-quality-attribution-odsc2026/blob/main/kg_diagnosis_adventureworks.ipynb

2. Choose **Runtime > Run all**. The first cell copies the tutorial repository and installs three small packages. Approve the Colab warning about running a notebook from GitHub.
3. Confirm the run ends with section 8, "The reveal", and no errors.
4. Optional: **File > Save a copy in Drive**, so your own copy is one click away during the session. Colab still needs a working internet connection while you use it.

Option B: your own Python environment (works fully offline once set up)

Requires Python 3.10 or newer and git. Run these commands once:

```
git clone https://github.com/Lokeshs-git/kg-quality-attribution-odsc2026.git
cd kg-quality-attribution-odsc2026
python -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install -r requirements.txt
jupyter lab kg_diagnosis_adventureworks.ipynb
```

Then choose **Run > Run All Cells**. After this first run, the notebook and its data are on your machine and the session needs no network at all.

Download sizes

What	Approximate size	When
Tutorial repository: notebook, data-prep script and AdventureWorks data as Parquet	4 MB	Both options
Python packages (pandas, NetworkX, DuckDB, LightGBM, SHAP, scikit-learn, matplotlib, JupyterLab)	275 MB	Option B only; less if you already have a scientific Python setup

Tools and languages

Python and SQL. Google Colab or JupyterLab; pandas, NetworkX, DuckDB, LightGBM, SHAP, scikit-learn and matplotlib. The dataset is Microsoft's AdventureWorks sample database, released under the MIT license.

If something goes wrong

If your environment will not run during the session, follow the walkthrough on screen. Every step is shown live, and you can run the notebook afterwards at your own pace. Repository: https://github.com/Lokeshs-git/kg-quality-attribution-odsc2026